

HW 5.1.1: Rational Functions

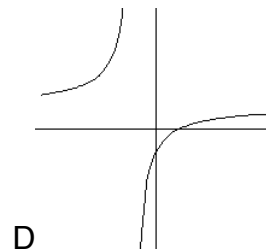
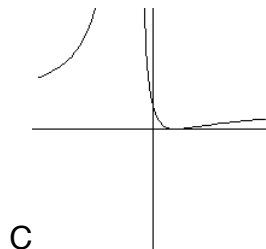
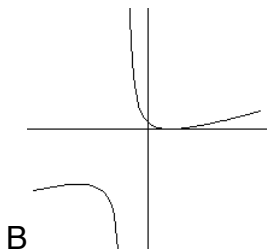
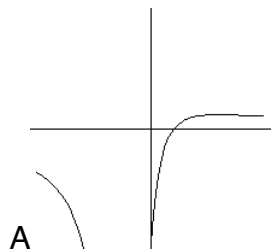
Match each equation form with one of the graphs. Assume $c > d$.

1. $f(x) = \frac{x-c}{x-d}$

2. $g(x) = \frac{(x-c)^2}{x-d}$

3. $h(x) = \frac{x-c}{(x-d)^2}$

4. $k(x) = \frac{(x-c)^2}{(x-d)^2}$



For each function, find the x-intercepts, y-intercept, vertical asymptotes, end behavior (horizontal asymptotes), and holes if applicable. Use that information to sketch the graph.

5. $p(x) = \frac{3x-5}{x+2}$

6. $q(x) = \frac{x-5}{3x-1}$

7. $s(x) = \frac{4}{(x-2)^2}$

8. $r(x) = \frac{6}{(x+2)^2}$

9. $f(x) = \frac{3x^2-14x-5}{3x^2+8x-16}$

10. $g(x) = \frac{2x^2-11x+15}{3x^2-14x+15}$

11. $a(x) = \frac{x^2 + 2x - 3}{x^2 - 1}$

12. $b(x) = \frac{x^2 + 4x - 5}{x^2 - 25}$

13. $h(x) = \frac{2x^2 + x - 1}{x^2 - 4}$

14. $k(x) = \frac{3x^2 - x - 10}{x^2 + 2x - 3}$

15. $n(x) = \frac{3x^2 + 4x - 4}{x^3 - 4x^2}$

16. $m(x) = \frac{5 - x}{2x^2 + 5x + 3}$

17. $w(x) = \frac{(x-1)(x+3)(x-5)}{(x+2)^2(x-4)}$

18. $z(x) = \frac{(x+1)^2(x-7)}{(x-5)(x+2)(x+3)}$

Write an equation for a rational function with the given characteristics.

19. Vertical asymptotes at $x = 5$ and $x = -5$

x intercepts at $(2,0)$ and $(-1,0)$

y intercept at $(0,4)$

20. Vertical asymptotes at $x = -3$ and $x = -2$

x intercepts at $(2,0)$ and $(1,0)$

y intercept at $(0,5)$

21. Vertical asymptotes at $x = -4$ and $x = -5$

x intercepts at $(4,0)$ and $(-6,0)$

Horizontal asymptote at $y = 7$

22. Vertical asymptotes at $x = -1$ and $x = 7$

x intercepts at $(-5,0)$ and $(2,0)$

Horizontal asymptote at $y = -3$

23. Vertical asymptote at $x = -1$

Double zero at $x = 2$

y intercept at $(0,2)$

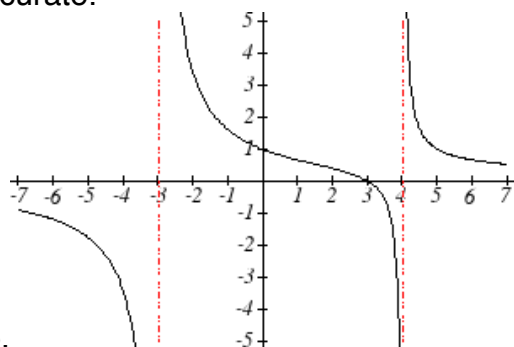
24. Vertical asymptote at $x = 4$

Double zero at $x = -1$

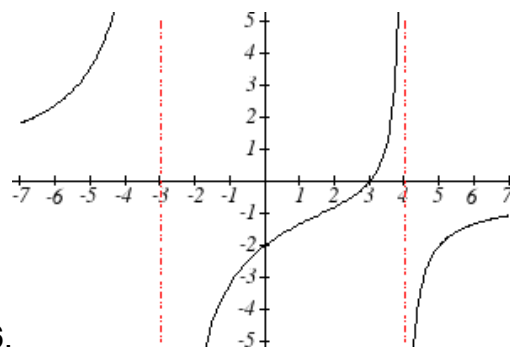
y intercept at $(0,3)$

Write a function for each graph. Answers will vary, but asymptote(s) and intercepts should be accurate.

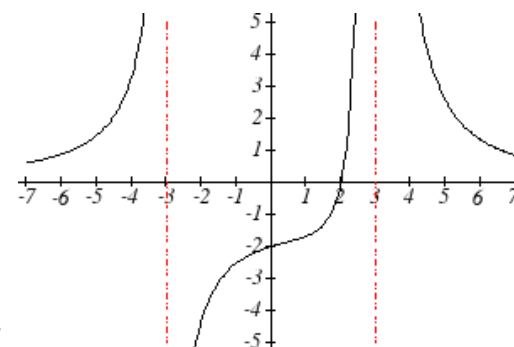
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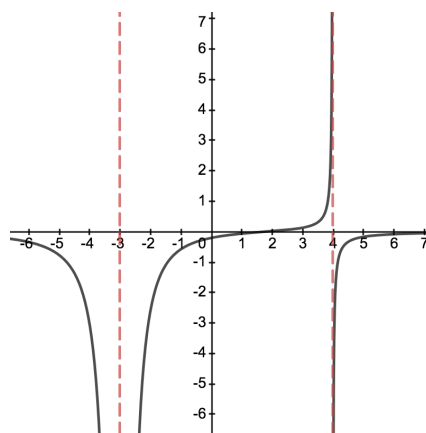
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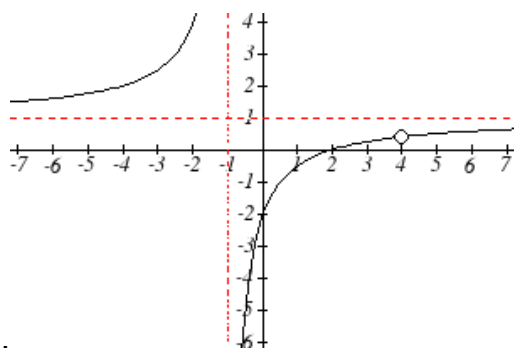
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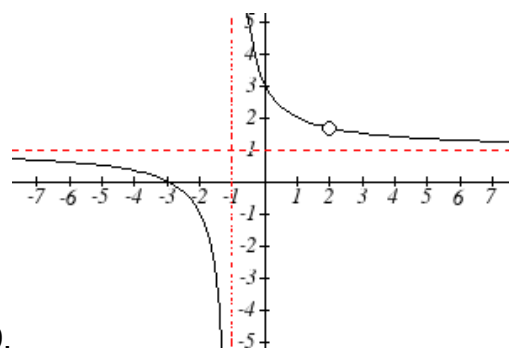
28.



29.



30.



Selected Answers:

1. D 2. B 3. A 4. C

5. Intercepts: $\left(\frac{5}{3}, 0\right), \left(0, -\frac{5}{2}\right)$ Vertical asymptotes: $x = -2$ Horizontal asymptotes: $y = 3$

Holes: none

7. Intercepts: $(0, 1)$ Vertical asymptotes: $x = 2$ Horizontal asymptotes: $y = 0$

Holes: none

9. Intercepts: $\left(-\frac{1}{3}, 0\right), (5, 0), \left(0, \frac{5}{16}\right)$ Vertical asymptotes: $x = -4, x = \frac{4}{3}$ Horizontal asymptotes: $y = 1$

Holes: none

11. Intercepts: $(-3, 0), (0, 3)$ Vertical asymptotes: $x = -1$ Horizontal asymptotes: $y = 1$ Holes: $x = 1$ 13. Intercepts: $\left(\frac{1}{2}, 0\right), (-1, 0), \left(0, \frac{1}{4}\right)$ Vertical asymptotes: $x = -2, x = 2$ Horizontal asymptotes: $y = 2$

Holes: none

15. Intercepts: $\left(\frac{2}{3}, 0\right), (-2, 0)$ Vertical asymptotes: $x = 0, x = 4$ Horizontal asymptotes: $y = 0$

17. Intercepts: $(1,0), (-3,0), (5,0), \left(0, -\frac{15}{16}\right)$

Vertical asymptotes: $x = -2, 4$

Horizontal asymptotes: $y = 1$

Holes: none

Functions will vary. Mainly looking for correct vertical asymptotes, horizontal asymptote, intercepts.

19. $y = \frac{50(x-2)(x+1)}{(x+5)(x-5)}$

21. $y = \frac{7(x-4)(x+6)}{(x+4)(x+5)}$

23. $y = \frac{(x-2)^2}{2(x+1)}$

25. $y = \frac{4(x-3)}{(x+3)(x-4)}$

27. $y = \frac{27(x-2)}{(x+3)(x-3)^2}$

29. $y = -\frac{(x-2)(x-4)}{(x-4)(x+1)}$